MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 7462

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, metal-ceramic
See figure 1
Mounting position: Any
Weight: 0.06 ounce (1.7 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	eb	Ec	egk	Ehk	Rk	Rg	Ik	Pp	TE	Barometric pressure, reduced
Unit:	V	Vdc	v	Vdc	Vdc	v	Ohms	Meg	mAdc	W	°C	mmHg
Maximum:	6.6	250	400	0,-50	0	50	---	0.01 (Note 1)	11.0	1.1	250 (Note 2)	8
Minimum:	6.0	---	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	150	---	+6	---	---	910	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓕ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
(F)	Qualification inspection								
1002	Barometric pressure, reduced	4	Pressure = 8 ±1 mmHg; voltage - 300 Vac	---	---	---	---	---	---
1031	Sweep-frequency vibration	9	Ebb = 150 Vdc; Rp - 10,000 ohms; Rk - 82 ohms; Ec = 0; 10 G; F = 100 to 2,000 Hz	---	---	Ep	---	15	mVac
1031	Sweep-frequency vibration fatigue	10	Eb = Ec = 0; Ef = 6.3 V	---	---	---	---	---	---
---	Sweep-frequency vibration fatigue test end points:								
1031	Low-frequency vibration	-		---	---	Ep	---	15	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	20	µAdc
1306	Transconductance (1)	-		---	---	Sm	6,800	---	µmhos
1301	Heater current	-		---	---	If	222	258	mA
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	222	258	mA
1336	Heater-cathode leakage	-		0.65	II	Ihk	---	20	µAdc
1306	Transconductance (1)	7		0.65	II	Sm	8,000	13,000	µmhos
1256	Electrode current (anode)	-	Ec = 0; Rk = 82 ohms	0.65	II	Ib	4.5	11.0	mAdc
1201	Short and discontinuity detection	-		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	-		---	---	R	100	---	Meg
1261	Electrode voltage (grid)	-	Rk = 0; Ec/Ib = 0.1 mAdc	---	---	Ec	---	-4.5	Vdc
1316	Amplification factor	7		---	---	Mu	65	115	---
1306	Transconductance (2)	7	Ef = 6.0 V	---	---	ΔSm Ef	---	15	%
1266	Grid emission	3	Ef = 7.0 V; Ecc = -10 Vdc; Rg = 0.1 Meg	---	---	Ic	0	-2.0	µAdc
1331	Direct-interelectrode capacitance	-		---	---	Cgp	1.05	1.45	pF
						Cin	1.25	2.25	pF
						Cout	0.013	0.045	pF
						Chk	1.1	1.9	pF
---	Envelope strain	6		---	---	---	---	---	---
(F) 1031	Low-frequency vibration	8	Ebb = 150 Vdc; Rp = 10,000 ohms; Rk = 82 ohms; Ec = 0:15 G; F = 40 Hz	---	---	Ep	---	10	mVac

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 3								
1506	Heater-cycling life	5	Ef = 7.0 V; Ehk = +70 Vdc; Rk = 0; Ec = Eb = 0	---	---	---	---	---	---
---	Heater-cycling life-test end point:								
1336	Heater-cathode leakage	-		---	---	Ihk	---	40	μAdc
1516	Stability life	-	Eb = 150 Vdc; Ecc = +6 Vdc; Ehk = -70 Vdc; Rk = 910 ohms; Rg = 0.1 Meg	2.5	Code F	---	---	---	---
---	Stability life-test end point (2 and 20 hours):								
1306	Change in transconductance (1) of individual tubes	-		---	---	ΔSm t	---	15	%
1501	Intermittent life	-	Group A: stability life-test conditions; t = 1,000 hours	---	---	---	---	---	---
---	Intermittent life-test end points:								
1301	Heater current	-		---	---	If	222	264	mA
1306	Transconductance (1)	-		---	---	Sm	6,500	---	μmhos
1306	Change in transconductance (1) of individual tubes	-		---	---	ΔSm t	---	30	%
1306	Transconductance (2)	-		---	---	ΔSm Ef	---	20	%
1336	Heater-cathode leakage	-		---	---	Ihk	---	20	μAdc
1211	Insulation of electrodes	-		---	---	R	50	---	Meg
(F) 1041	Shock	11	Hammer angle = 30°; Ehk = 50 Vdc; Eb = 150 Vdc; Rk = 82 ohms; Ec = 0	---	---	---	---	---	---
---	Shock test end points:								
1031	Low-frequency vibration	-		---	---	Ep	---	15	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	20	μAdc
1306	Transconductance (1)	-		---	---	Sm	6,800	---	μmhos
1301	Heater current	-		---	---	If	222	258	mA

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NOTES:

1. If resistance is used in the cathode or anode circuits, the grid circuit resistance may be as high as (10,000 + 100 Rk + 10 Rℓ) ohms, where Rk is the cathode-bias resistance in ohms and the Rℓ is the ac anode load resistance in ohms.

2. Operation below the rated maximum temperature is recommended for applications requiring the longest possible tube life.

3. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions specified below. The 3-minute test is not permitted. Test within 3 seconds after preheating.

Ef V	Ecc Vdc	Eb Vdc	Rk Ohms	Rg Meg
7.0	0	150	82	0.1

4. The specified voltages shall be applied between the anode and grid.

5. Sampling and acceptance criteria shall be as follows: First sample (n1) shall consist of 15 tubes with an acceptance number (c1) of zero. Second sample (n2) shall consist of 15 tubes with an acceptance number (cc) equal to one for the cumulative sample.

6. Tubes shall be tested as specified in method 2126, except that they shall be first immersed in water at not more than 5 °C for 15 seconds and immediately thereafter subjected to the standard temperature cycle specified in method 2126. Tubes having high-heater current of 300 mA, or more, shall be rejected as evidence of air leaks.

7. In performing this test, the cathode by-pass capacitor should be charged to + 6 volts prior to insertion of the tube into the test socket to prevent high-initial surge current.

8. Low-frequency vibration. Test shall be performed as follows: The tube shall be vibrated with simple harmonic motion in each of two planes; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and perpendicular to major axis of terminal lug.

9. Sweep-frequency vibration. Test shall be performed as follows:

   (a) The frequency shall be increased from 100 to 2,000 Hz with approximately logarithmic progression in 3 ±1 minutes. The return sweep (2,000 to 100 Hz) is not required.
   (b) The tube shall be vibrated with simple harmonic motion in each of two planes; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and parallel to a line through the major axis of a terminal lug. At all frequencies from 100 to 2,000 Hz, the total harmonic distortion of the acceleration waveform shall be less than 5 percent.
   (c) The peak acceleration shall be maintained at 10 ±1.0 G throughout the test.
   (c) The value of the alternating voltage, Ep, produced across the resistor, Rp, as a result of the vibration shall be measured with a suitable device having a response of the RMS value of the voltage to within ±0.5 dB of the response at 400 Hz for the frequency range of 100 to 3,000 Hz, and having a bandpass filter with an attenuation rate of 24 dB per octave below the low-frequency cutoff point of 50 Hz and above the high-frequency cutoff point of 5,000 Hz. The meter shall have a dynamic response characteristic equivalent to, or faster than, a VU meter (operated in accordance with Standard ASA No. C16.5-1954).

10. Sweep-frequency vibration fatigue. Test shall be performed as specified in method 1031, except that the tubes shall be vibrated for a total of 6 hours, that is, 3 hours in each of two directions; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and parallel to a line through the major axis of a terminal lug.

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NOTES: -Continued

(F) 11. This test shall be performed during the initial production and one each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU
Air Force - 80
DSA - ES

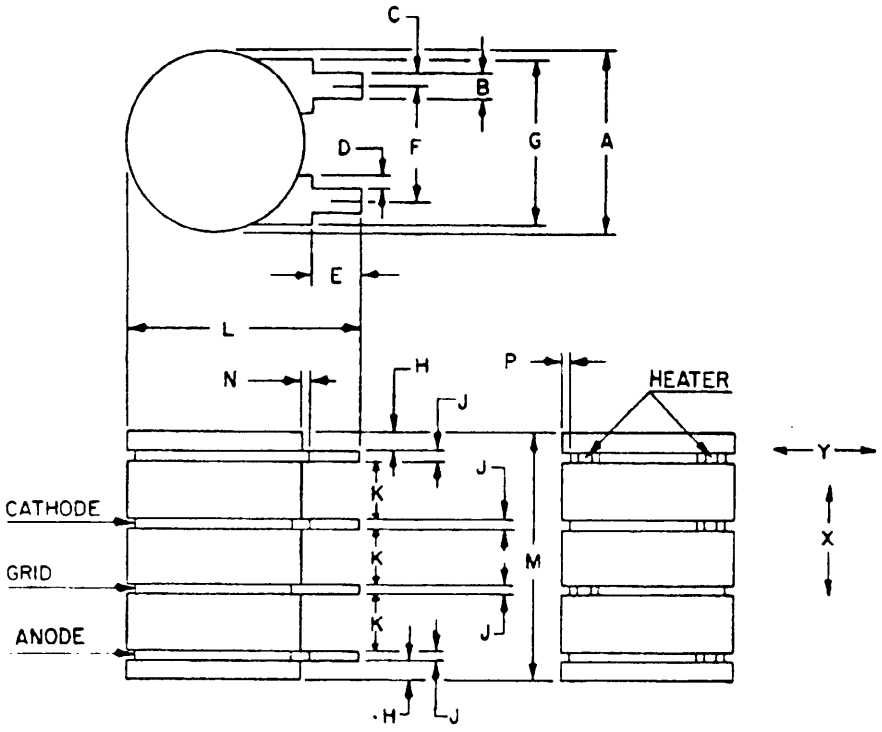
User activities:
Army - ME, WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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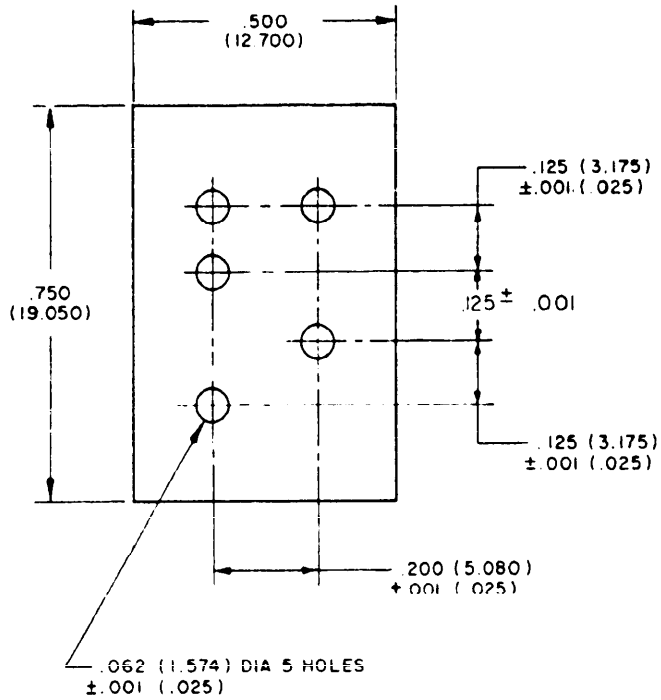


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	.316 (8.03)	.330 (8.38)
B	.046 (1.17)	.052 (1.32)
E	.096 (2.44)	.104 (2.64)
J	.020 (.51)	.026 (.66)
L	.429(10.90)	.459(11.66)
M	.460(11.68)	.490(12.45)
Quality conformance inspection, part 2		
C	.023 (.58)	.031 (.79)
D	.018 (.46)	.026 (.66)
F	.190 (4.83)	.210 (5.33)
K	.100 (2.54)	.110 (2.79)
Reference dimensions		
G	.300(7.62)	
H	.040(1.02)	
N	.020 (.51)	
P	.012 (.30)	

NOTE:
All terminals shall be so alined that they are capable of being inserted into a gage 1/8 inch (3.05 mm) thick having 5 holes .062(1.57 mm) ±.001(.03 mm) in diameter as shown on figure 2.

FIGURE 1. Outline drawing of electron tube type 7462.

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NOTE:
Dimensions are in inches (millimeters are in parentheses).

FIGURE 2. Terminal position gage.

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NAVAL ELECTRONIC SYSTEMS COMMAND
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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